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Intelligent Cyber Threat Prediction Model for Blockchain-Enabled Land Revenue Records

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Introduction

- Land revenue records play a crucial role in property ownership, legal verification, and government administration. Traditionally, these records were maintained using paper-based or centralized digital systems, which are prone to limitations such as **data tampering, lack of transparency, delays in access, and vulnerability to cyber attacks**.
- With the advancement of digital technologies, many governments have shifted toward online land record management systems, improving accessibility and efficiency. However, this shift also introduces challenges related to **data security, privacy, and unauthorized access**. Cyber threats such as hacking, data breaches, and manipulation of records can lead to serious legal and financial consequences.
- Therefore, there is a need for a **secure and intelligent system** to protect sensitive land records and ensure reliable data management.

Objective

- To design and develop an **intelligent system** for secure land revenue record management
- To integrate **Blockchain technology** for **decentralized, transparent, and tamper-proof storage**
- To implement **Machine Learning algorithms** (Random Forest, KNN) for **analyzing system behavior**
- To **detect anomalies and predict cyber threats** at an early stage To provide **real-time monitoring** of system activities
- To generate **instant alerts and notifications** to administrators
- To ensure **data integrity, privacy, and reliability**
- To reduce risks of **data breaches, fraud, and unauthorized access**
- To build a system suitable for **real-time government applications**

Problem Statement

- Existing land record systems are mostly **centralized**, making them highly vulnerable to **cyber attacks and system failures**
- Increasing incidents of **data tampering, fraud, and unauthorized access** threaten the authenticity of records
- Lack of **transparency and traceability**, making it difficult to verify changes in data
- No effective **real-time monitoring system** to detect suspicious activities
- Absence of **early threat prediction mechanisms** in current solutions
- Difficulty in ensuring **data integrity, privacy, and security**
- Traditional security methods are **not sufficient** to handle modern cyber threats
- Growing need for a **secure, reliable, and intelligent system** to protect sensitive land records

Literature Survey (10 papers)

Author/Year	Title	Methodology	Advantages and Disadvantages
Sharma et al., 2026	AI-Based Proactive Threat Detection for Government Land Records	Machine learning anomaly detection applied to land revenue databases to identify suspicious access patterns before attacks occur.	Adv: Early threat detection and improved protection of land records. Dis: Requires large datasets and continuous model training.
Patel & Singh, 2025	Blockchain-Secured Land Registry with Security Monitoring	Blockchain used for immutable storage of property records combined with monitoring tools for system activity.	Adv: High transparency and tamper resistance. Dis: Lacks proactive cyber-attack prediction capability.
Kumar et al., 2024	Secure Cloud Framework for Digital Land Revenue Systems	Cloud-based architecture with encryption and access control for land record management systems.	Adv: Scalable storage and secure access. Dis: Insider threat detection is limited.
Rahman & Das, 2023	Cybersecurity Challenges in E-Governance Land Information Systems	Analysis of security policies and vulnerabilities in government land information infrastructures.	Adv: Identifies critical security gaps in e-governance platforms. Dis: Mostly theoretical without automated detection.

Chen et al., 2022	Machine Learning Models for Government Database Security	Classification algorithms used to detect abnormal user behavior in government databases.	Adv: Improves detection of unusual access patterns. Dis: High computational requirements.
Nair & Reddy, 2021	Digitized Land Registry System with Security Framework	Development of a digital platform for land record storage with authentication and encryption mechanisms.	Adv: Enhances transparency and digital accessibility. Dis: Security remains reactive rather than predictive.
Singh et al., 2020	Intrusion Detection System for Government Databases	Network-based IDS monitors system traffic to detect unauthorized access attempts.	Adv: Effective in detecting active intrusions. Dis: Cannot predict malicious intent before attacks.

Zhang et al., 2019	Blockchain-Based Protection for Property Records	Distributed ledger technology ensures tamper-proof storage of land ownership information.	Adv: High data integrity and transparency. Dis: Scalability and performance challenges.
Verma & Gupta, 2018	Risk Assessment in Land Information Systems	Cyber risk analysis methodology used to evaluate vulnerabilities in digital land systems.	Adv: Helps identify major security risks. Dis: No automated prevention or detection system.
Alam et al., 2017	Security Issues in Digital Land Record Management	Comprehensive study of vulnerabilities and threats affecting digital land record infrastructures.	Adv: Highlights importance of securing government property data. Dis: Does not provide proactive defense mechanisms.

Requirements

1. Hardware Requirements

- **Computer / Laptop:**
  A system capable of running development tools and applications efficiently.
- **Processor:**
  Minimum **Intel i5 or above** for smooth processing of machine learning tasks.
- **RAM:**
  At least **8 GB RAM** to handle data processing and model execution.
- **Storage:**
  Minimum **256 GB hard disk** for storing datasets, models, and application files.
- **Internet Connection:**
  Stable connection required for blockchain operations and data access.

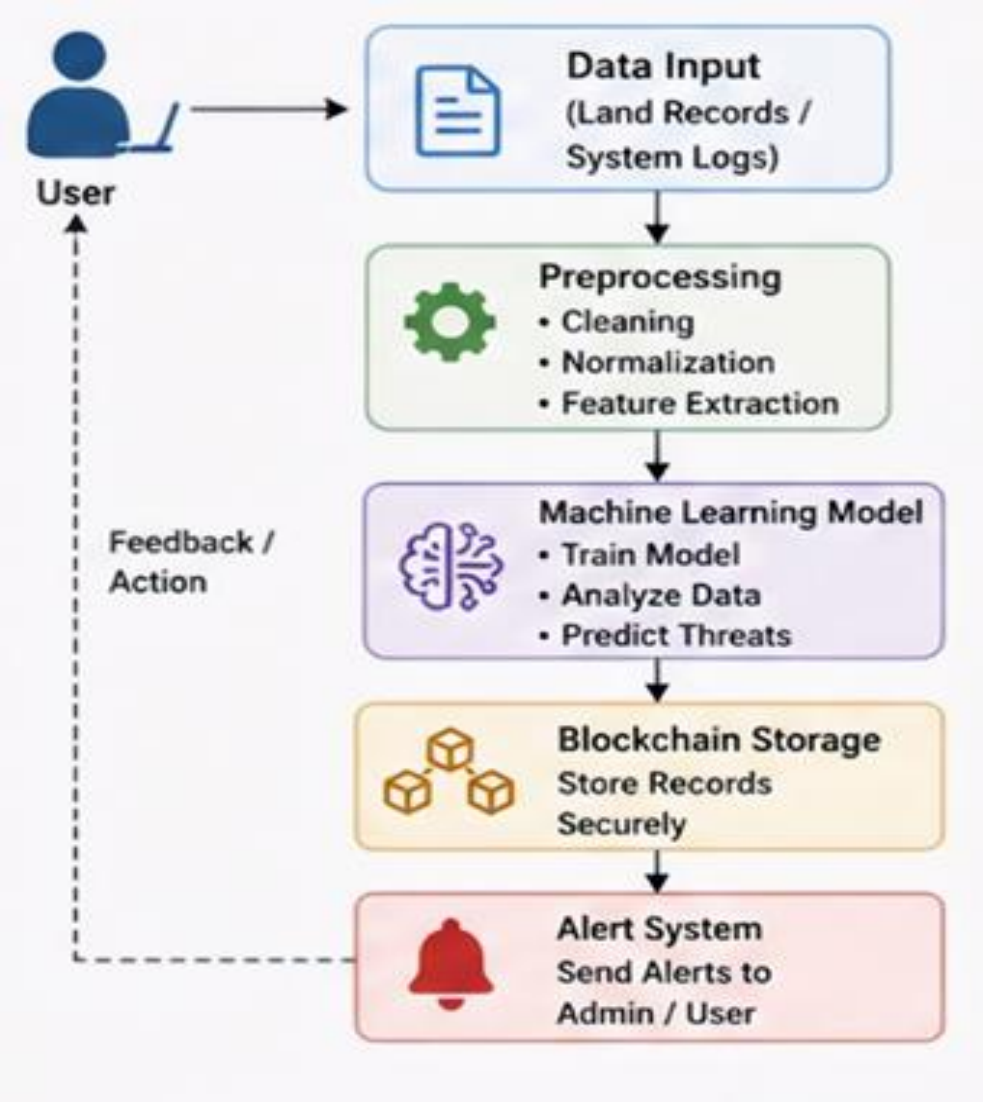
2. Software Requirements

- **Programming Language:**
  Python for implementing machine learning models and backend logic
- **Machine Learning Libraries:**
  Scikit-learn and TensorFlow for training and testing models
- **Blockchain Platform:**
  Ethereum or Hyperledger for secure and decentralized data storage
- **Database:**
  MySQL or MongoDB for managing and storing application data
- **IDE / Development Tools:**
  VS Code, PyCharm, or Jupyter Notebook for development and testing
- **Operating System:**
  Windows or Linux for running the application

Methodology

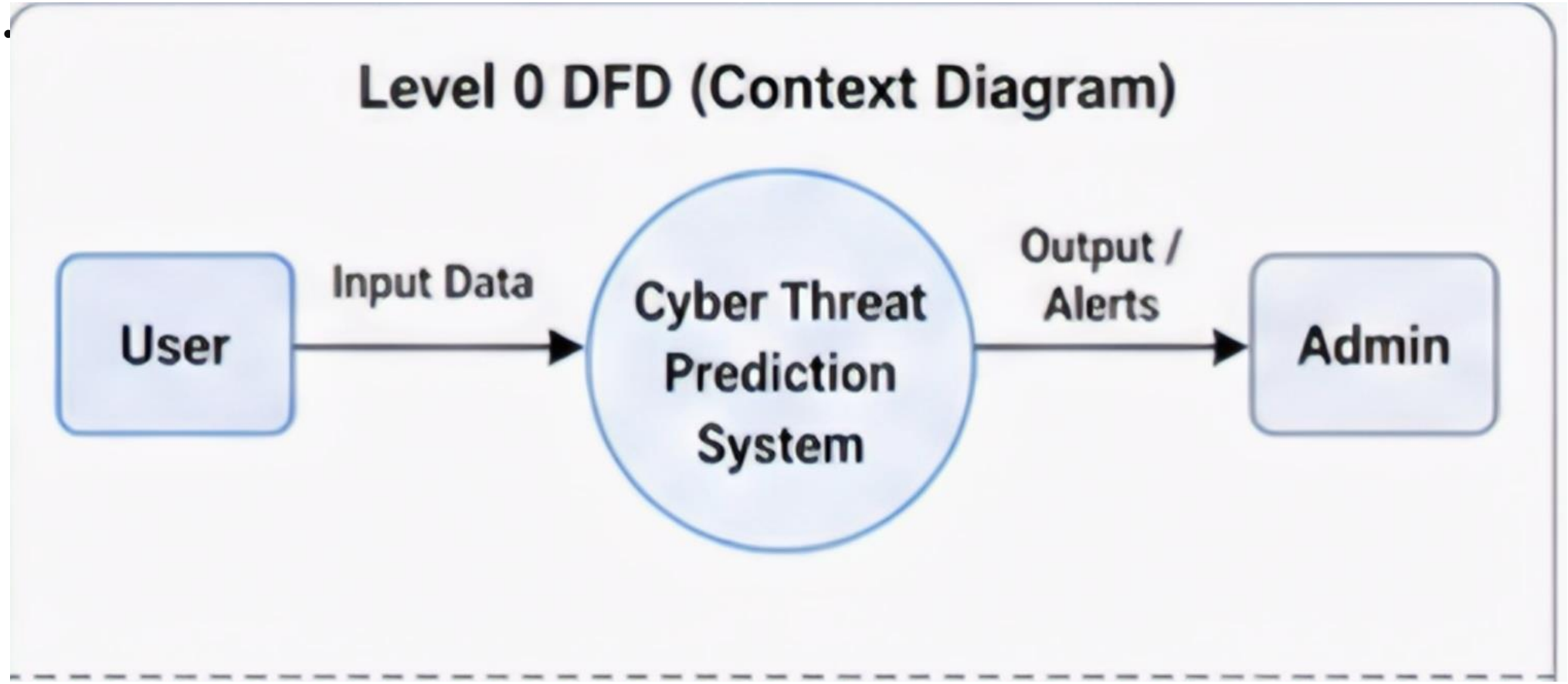
- **Data Collection:**
  Collect cyber attack datasets and land record data
- **Data Preprocessing:**
  Clean data, remove noise, and handle missing values
- **Feature Extraction:**
  Identify important features related to cyber threats
- **Model Training:**
  Train Machine Learning models (Random Forest, KNN)
- **Model Testing:**
  Evaluate model accuracy and performance
- **Threat Prediction:**
  Detect anomalies and predict cyber attacks
- **Blockchain Integration:**
  Store land records securely using blockchain
- **Alert Generation:**
  Generate real-time alerts for suspicious activities

System Architecture

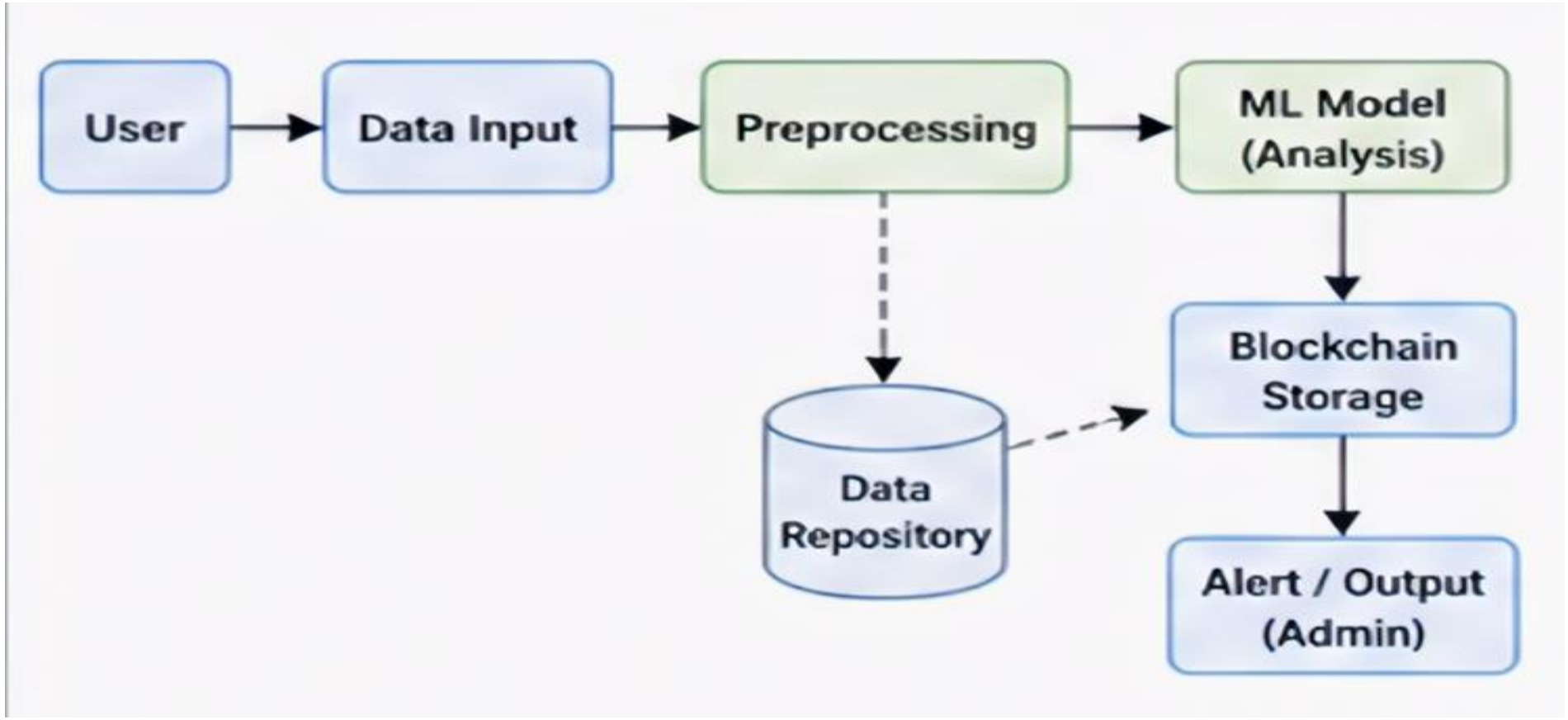


- **User Interface Layer:**
  Allows users to **upload, view, and manage land records** through a web-based interface.
- **Data Input Module:**
  Collects land record details and system activity data for further processing.
- **Data Preprocessing Module:**
  Cleans and processes the data by removing noise, handling missing values, and preparing it for analysis.
- **Machine Learning Module:**
  Applies algorithms like **Random Forest and KNN** to analyze data and detect anomalies or potential cyber threats.
- **Blockchain Storage Layer:**
  Stores land records in a **secure, decentralized, and tamper-proof** manner using blockchain technology.
- **Alert and Monitoring System:**
  Generates **real-time alerts** when suspicious activities or threats are detected.
- **Admin Dashboard:**
  Allows administrators to **monitor system activity, view alerts, and take necessary actions**.

Data Flow Diagram



Data Flow Diagram



Effective Technique

Techniques Used

- Blockchain Technology
- Random Forest Algorithm
- K-Nearest Neighbors (KNN)

1.Why These Techniques Are Chosen

- **Blockchain:**
  Ensures **secure, decentralized, and tamper-proof storage** of land records
- **Random Forest:**
  Provides **high accuracy**, handles large datasets, and reduces overfitting
- **KNN:**
  Simple and effective for **detecting anomalies based on data similarity**

2.Why Better Than Other Techniques

- **Blockchain vs Traditional Database:**
  More secure and **prevents data tampering**
- **Random Forest vs Decision Tree:**
  More accurate and **less prone to overfitting**
- **KNN vs Complex Models:**
  Easier to implement and effective for **pattern-based detection**

3.How It Matches Project Objectives

- **Security & Data Integrity** → Achieved using Blockchain
- **Threat Detection** → Achieved using ML algorithms
- **Real-time Monitoring** → Enabled through anomaly detection
- **Reliability & Transparency** → Improved through decentralized system

Conclusion

The proposed system provides a **secure and efficient solution** for managing land revenue records. By integrating **Blockchain technology** and **Machine Learning algorithms**, the system ensures **data integrity, transparency, and protection against cyber threats**. The machine learning model enables **early detection of suspicious activities**, while blockchain guarantees **tamper-proof storage** of records. This approach significantly reduces the risks of **data breaches, unauthorized access, and fraud**, thereby improving the overall **reliability and trust** in land record management systems.

Machine learning techniques enable **early detection and prediction of cyber threats** by analyzing patterns and identifying anomalies in system behavior. This allows the system to respond quickly to suspicious activities and prevent potential attacks.

By combining these technologies, the system significantly reduces the risks of **data breaches, fraud, and unauthorized access**, while ensuring **data integrity, privacy, and reliability**. It also supports **real-time monitoring and alert generation**, improving overall system efficiency.

Thus, the proposed model offers a **powerful and future-ready solution** for secure land record management and can be effectively applied in **real-world government systems**.

Future Enhancement

- **Real-Time Implementation:**
  Deploy the system in **government applications** for live monitoring of land records.
- **Advanced AI Models:**
  Use **deep learning techniques** to improve accuracy of threat prediction.
- **Mobile Application:**
  Develop a mobile app for **easy access and monitoring** by users and administrators.
- **Integration with Government Systems:**
  Link with systems like **digital identity services (e.g., Aadhaar)** for better verification.
- **Enhanced Security Features:**
  Implement **multi-factor authentication and encryption** for additional protection.

THANK YOU